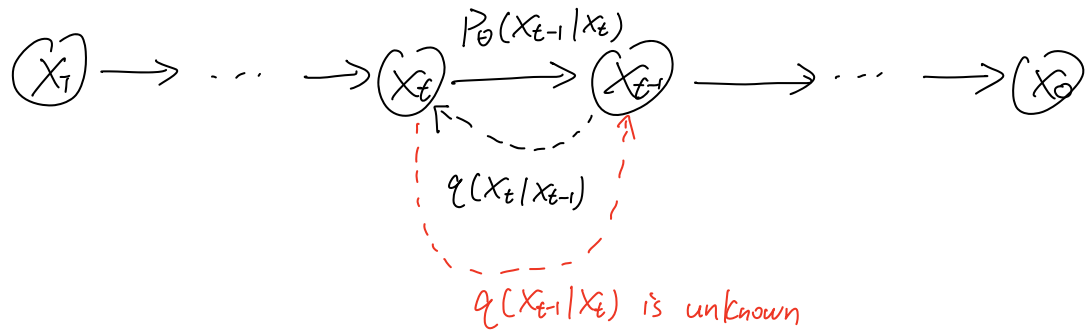


Denoising Diffusion Probabilistic Models.

Graphical Model of DDPM.



Model

sequence from x_0 to x_T

$$p_\theta(x_0) := \int p_\theta(x_{0:T}) d\textcircled{x_{1:T}} \text{ --- latent}$$

Forward Process:

$$q(x_{1:T} | x_0) := \prod_{t=1}^T q(x_t | x_{t-1}), \quad q(x_t | x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I})$$

β_t could be trainable
but fixed here...

Reverse Process:

$$p(x_T) = \mathcal{N}(x_T; 0, \mathbf{I})$$

$$p_\theta(x_{0:T}) := p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t), \quad p_\theta(x_{t-1} | x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

Loss (ELBO)

$$-\log p_{\theta}(x_0) \leq -\log p_{\theta}(x_0) + \text{KL}(q(x_{1:T}|x_0) \parallel p_{\theta}(x_{1:T}|x_0))$$

$$= -\log p_{\theta}(x_0) + \mathbb{E}_{x_{1:T} \sim q(x_{1:T}|x_0)} \left[\log \frac{q(x_{1:T}|x_0)}{p_{\theta}(x_{0:T})/p_{\theta}(x_0)} \right]$$

$$= -\log p_{\theta}(x_0) + \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{q(x_{1:T}|x_0)}{p_{\theta}(x_{0:T})} + \log p_{\theta}(x_0) \right]$$

$$= \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{q(x_{1:T}|x_0)}{p_{\theta}(x_{0:T})} \right]$$

$$\text{let } L = \mathbb{E}_{q(x_{0:T})} \left[\log \frac{q(x_{1:T}|x_0)}{p_{\theta}(x_{0:T})} \right]$$

$$= \mathbb{E}_{q(x_0)} \left[\mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{q(x_{1:T}|x_0)}{p_{\theta}(x_{0:T})} \right] \right] \geq -\mathbb{E}_{\boxed{q(x_0)}} \log p_{\theta}(x_0)$$

data distribution?

Reparameterization Trick.

We already have $q(x_t | x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \bar{I})$

Let $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$, we have

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon_{t-1} \quad \text{where } \epsilon_{t-1}, \epsilon_{t-2}, \dots \sim \mathcal{N}(0, \bar{I})$$

$$= \sqrt{\alpha_t} \sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{\alpha_t} \sqrt{1 - \alpha_{t-1}} \epsilon_{t-2} + \sqrt{1 - \alpha_t} \epsilon_{t-1}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1}) + 1 - \alpha_t} \bar{\epsilon}_{t-2} \quad \text{where } \bar{\epsilon}_{t-2} \text{ merges two Gaussians}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\epsilon}_{t-2}$$

$$= \dots$$

$$= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

$$A \sim \mathcal{N}(0, \sigma_A^2), B \sim \mathcal{N}(0, \sigma_B^2)$$

$$\text{then } A+B \sim \mathcal{N}(0, \sigma_A^2 + \sigma_B^2)$$

then we get a "nice property": $q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \bar{I})$.

Try to make loss tractable.

$$L = \mathbb{E}_{q(x_{0:T})} \left[-\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)} \right]$$

$$= \mathbb{E}_{q(x_{0:T})} \left[-\log p(x_T) - \sum_{t \geq 1} \log \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})} \right]$$

= ... Appendix A of the DDPM paper.

$$= \mathbb{E}_{q(x_{0:T})} \left[\underbrace{\text{KL}(q(x_T|x_0) \parallel p(x_T))}_{\text{Gaussian}} + \sum_{t \geq 1} \left[\underbrace{\text{KL}(q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t))}_{\text{unknown.}} - \underbrace{\log p_\theta(x_0|x_t)}_{\text{Gaussian}} \right] \right]$$

Good news: It is Gaussian!

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$$

$$\text{where } \tilde{\mu}_t(x_t, x_0) := \frac{\sqrt{\alpha_{t+1}} \beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t \quad \text{and} \quad \tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

The choice of θ

We have $p_\theta(X_{0:T}) := p(X_T) \prod_{t=1}^T p_\theta(X_{t-1}|X_t)$, $p_\theta(X_{t-1}|X_t) := \mathcal{N}(X_{t-1}; \mu_\theta(X_t, t), \Sigma_\theta(X_t, t))$

For variance, we set $\Sigma_\theta(X_t, t) = \sigma_t^2 I$ where $\sigma_t^2 = \beta_t$ or $\sigma_t^2 = \tilde{\beta}_t = \frac{1 - \bar{\alpha}_t}{1 - \bar{\alpha}_1} \beta_t$.

For mean $\mu_\theta(X_t, t)$, firstly, we recall the loss L

$$L = \mathbb{E}_{q(X_{0:T})} \left[\underbrace{KL(q(X_T|X_0) || p(X_T))}_{L_T} + \sum_{t=1}^T \underbrace{KL(q(X_{t-1}|X_t, X_0) || p_\theta(X_{t-1}|X_t))}_{L_{t-1}} - \underbrace{\log p_\theta(X_0|X_1)}_{L_0} \right]$$

For L_T , since we regard β_t as constants, then

$q(X_t|X_0) = \mathcal{N}(X_t; \sqrt{\alpha_t} X_0, (1 - \bar{\alpha}_t)I)$ does not have any trainable args.

so L_T is a constant.

$$\mathcal{L} = \mathbb{E}_{q(x_{0:T})} \left[\underbrace{\text{KL}(q(x_T|x_0) || p(x_T))}_{\mathcal{L}_T} + \sum_{t=1}^T \underbrace{\text{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t))}_{\mathcal{L}_{t-1}} - \underbrace{\log p_\theta(x_0|x_1)}_{\mathcal{L}_0} \right]$$

For $\mathcal{L}_{t-1}$, recall

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$$

$$\text{where } \tilde{\mu}_t(x_t, x_0) := \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t \quad \text{and} \quad \tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

$$\text{and} \quad p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I) \quad \text{let } \sigma_t^2 = \tilde{\beta}_t$$

$$\text{we have } \mathcal{L}_{t-1} = \mathbb{E}_{q(x_{0:T})} \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}(x_t, x_0) - \mu_\theta(x_t, t)\|^2 \right] + C \quad \text{const}$$

Recall the reparameterization trick $X_t(X_0, \epsilon) = \sqrt{\alpha_t} X_0 + \sqrt{1-\alpha_t} \epsilon$

then

$(\epsilon \sim \mathcal{N}(0, I))$

$$\begin{aligned} \mathcal{L}_{t+1} - \mathcal{L} &= \mathbb{E}_{X_0, \epsilon} \left[\frac{1}{2\sigma_t^2} \left\| \tilde{\mu}_t \left(X_t(X_0, \epsilon), \frac{1}{\sqrt{\alpha_t}} (X_t(X_0, \epsilon) - \sqrt{1-\alpha_t} \epsilon) \right) - \mu_\theta(X_t(X_0, \epsilon), t) \right\|^2 \right] \\ &= \mathbb{E}_{X_0, \epsilon} \left[\frac{1}{2\sigma_t^2} \left\| \frac{1}{\sqrt{\alpha_t}} (X_t(X_0, \epsilon) - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon) - \mu_\theta(X_t(X_0, \epsilon), t) \right\|^2 \right] \quad (10) \end{aligned}$$

eq (10) reveals that μ_θ must predict $\frac{1}{\sqrt{\alpha_t}} (X_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon)$ given X_t

so we choose $\hat{\epsilon}$ that

$$\mu_\theta(X_t, t) = \tilde{\mu}_t \left(X_t, \frac{1}{\sqrt{\alpha_t}} (X_t - \sqrt{1-\alpha_t} \epsilon_\theta(X_t)) \right) = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(X_t, t) \right)$$

where ϵ_θ is a function approximator intended to predict ϵ from X_t

Hence, to sample $X_{t-1} \sim p_\theta(X_{t-1}|X_t)$ is to compute

$$X_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(X_t, t) \right) + \epsilon_t Z, \quad Z \sim \mathcal{N}(0, I)$$

Then (10) is simplified to

$$L_{t+1} - C = \mathbb{E}_{x_t, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1-\bar{\alpha}_t)} \| \epsilon - \epsilon_\theta (\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} \epsilon, t) \|^2 \right].$$

In fact, Ho's work indicate that

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t, x_0, \epsilon} [\| \epsilon - \epsilon_\theta (\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t} \epsilon, t) \|^2] \text{ works better}$$

Table 2: Unconditional CIFAR10 reverse process parameterization and training objective ablation. Blank entries were unstable to train and generated poor samples with out-of-range scores.

Objective	IS	FID
$\tilde{\mu}$ prediction (baseline)		
L , learned diagonal Σ	7.28 ± 0.10	23.69
L , fixed isotropic Σ	8.06 ± 0.09	13.22
$\ \tilde{\mu} - \tilde{\mu}_\theta\ ^2$	–	–
ϵ prediction (ours)		
L , learned diagonal Σ	–	–
L , fixed isotropic Σ	7.67 ± 0.13	13.51
$\ \tilde{\epsilon} - \epsilon_\theta\ ^2 (L_{\text{simple}})$	9.46 ± 0.11	3.17

Generation Result.

Table 1: CIFAR10 results. NLL measured in bits/dim.

Model	IS	FID	NLL Test (Train)
Conditional			
EBM [11]	8.30	37.9	
JEM [17]	8.76	38.4	
BigGAN [3]	9.22	14.73	
StyleGAN2 + ADA (v1) [29]	10.06	2.67	
Unconditional			
Diffusion (original) [53]			≤ 5.40
Gated PixelCNN [59]	4.60	65.93	3.03 (2.90)
Sparse Transformer [7]			2.80
PixelIQN [43]	5.29	49.46	
EBM [11]	6.78	38.2	
NCSNv2 [56]		31.75	
NCSN [55]	8.87 ± 0.12	25.32	
SNGAN [39]	8.22 ± 0.05	21.7	
SNGAN-DDLS [4]	9.09 ± 0.10	15.42	
StyleGAN2 + ADA (v1) [29]	9.74 ± 0.05	3.26	
Ours (L , fixed isotropic Σ)	7.67 ± 0.13	13.51	≤ 3.70 (3.69)
Ours (L_{simple})	9.46 ± 0.11	3.17	≤ 3.75 (3.72)